US SIGNAL CREATED ARCHITECTURE DOCUMENT

Celar AI Private Intelligence Architecture

Celeritas: speed of light. Celar AI: speed of delivery. Private-first intelligence for unified solution-provider operations

Document owner	Product	Version	Classification
US Signal	Pulse	2.0	US Signal Internal — Confidential
Architecture status	Module	Prepared for	Canonical name
Review baseline	011 — Celar AI	US Signal architecture review	Pulse

This architecture package is created and owned by US Signal. Celar AI was conceived and engineered under the direction of Dr. Ahmed Adeyemi, Manager of Professional Services, and uses the supplied US Signal logo.

Document Control

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Contents

1. Executive Summary

Celar AI is the unified operational intelligence system for the US Signal Solution Provider division. It was conceived and engineered under the direction of Dr. Ahmed Adeyemi, Manager of Professional Services, as the central intersection where consulting teams convene, collaborate, and exchange critical project information.

The name draws from celeritas—the Latin concept of swiftness or speed—and from the conventional symbol c for the speed of light in $E=mc^2$. This connection honors US Signal's fiber-network heritage while defining the Professional Services mission to translate the speed of light into the speed of delivery.

Celar AI emerged from the operational friction experienced with legacy Changepoint workflows: siloed information, rigid navigation, repetitive administration, slow SOW preparation, fragmented task handoffs, time-entry burden, and delayed financial visibility.

From a solution-provider perspective, Celar AI unifies authorized documents, live system data, delivery workflows, troubleshooting evidence, reports, and financial context so teams can scope, execute, troubleshoot, report, and invoice work faster without abandoning security, governance, source-system ownership, or human accountability.

Celar AI is the private, permission-aware intelligence layer for Pulse. It combines authorized internal documents, governed live system data, deterministic calculations, and a private language model so users receive detailed, source-grounded answers and reviewable drafts without exposing confidential information to public external models.

Every request begins with authentication and authorization. Pulse retrieves only the document sections and live records the effective user is allowed to access. The private Celar AI model reasons over that context, produces a structured result, and measures confidence, source coverage, freshness, conflicts, and calculation validity.

When private evidence is sufficient, the result proceeds directly to private verification. When generic reasoning could improve the response and policy allows it, Pulse creates a minimal sanitized problem capsule, routes it through Module 064 to an approved external provider, and re-verifies the external result against private authoritative evidence before returning it to the user.

2. Scope, Objectives, and Outcomes

In scope are Pulse web and API services; identity, RBAC and View-As; project, customer, team, record, and field scope; private document retrieval; governed read-only tools; private embeddings and vector search; private model inference; confidence assessment; evidence verification; Module 064 controlled egress; DLP, audit, evaluation, model registry, and training governance.

Out of scope are autonomous approval or submission, unrestricted model-generated SQL, direct model access to production credentials, raw internal-document transmission to public external models, and autonomous changes to prompts, permissions, model weights, deployments, or production routes.

The target outcome is one reusable intelligence fabric for Timesheet, Help and Search, FlowHive, reports, financial analysis, and future authorized Pulse use cases.

3. Architecture Principles and Decisions

- Private first: restricted data is retrieved and reasoned over inside the approved Pulse trust boundary.
- Authorization before retrieval: the application—not the model—decides which data may be accessed.
- Facts from tools, explanation from models: deterministic services own calculations, permissions, schedules, and record state.
- Progressive disclosure: responses lead with a direct conclusion and then expose evidence, calculations, assumptions, conflicts, limitations, risks, and next actions.
- External models are optional: Claude or OpenAI may provide generic reasoning only after DLP sanitization and policy approval.
- Human control: timesheets, project baselines, financial actions, datasets, model versions, and production promotion remain human reviewed.

4. Logical Architecture

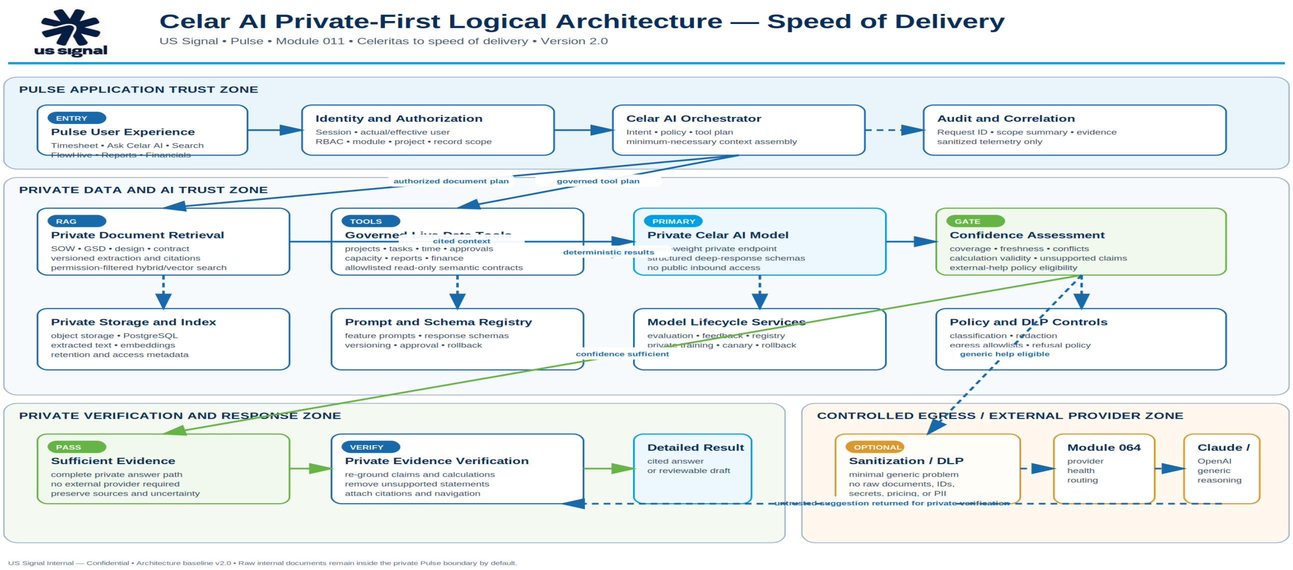

Figure 1. Celar AI private-first logical architecture.

The logical design separates the Pulse application trust zone, private data and AI zones, controlled egress, and external-provider boundary. Solid lines represent primary operational flow. Dashed lines represent optional sanitized reasoning, telemetry, or audit flow.

The confidence service is the decision point between a complete private answer and optional generic external assistance. External assistance never bypasses private evidence verification.

5. Component Architecture

Pulse experience layer: Timesheet, Ask Celar AI, global Search, FlowHive, Reporting, Financial Insights, and future module surfaces.

Identity and policy layer: session validation, actual/effective user resolution, module/action permissions, project and customer scope, field restrictions, and data classification.

Knowledge layer: malware-scanned private storage, document versioning, extraction, OCR when required, classification, chunking, embeddings, and permission-filtered hybrid retrieval.

Tool layer: allowlisted read-only module APIs and semantic metric contracts for projects, time, approvals, utilization, capacity, reports, contracts, rates, expenses, billing, invoices, operations, and audit.

Reasoning layer: private model endpoint, prompt and schema registry, confidence service, citation builder, response composer, private verifier, and governed local fallback.

External control layer: sanitization and DLP gateway, Module 064 provider configuration and routing, external provider adapters, circuit breakers, and sanitized telemetry.

Layer	Key components	Authority
Experience	Timesheet, Ask Celar AI, Search, FlowHive, Reports	Owning modules
Policy	Identity, RBAC, scope, classification	Pulse backend
Knowledge	Storage, extraction, embeddings, search	Private data services
Tools	Read-only APIs and metrics	Owning modules
Reasoning	Private model, confidence, verifier	Module 011
Egress	DLP, Module 064, external providers	Security + Module 064

6. Trust Zones and Data Boundaries

Zone A — User and edge: browser, identity provider, WAF/reverse proxy, and authenticated Pulse session.

Zone B — Pulse application: web/API services, authorization, orchestration, module tools, and audit correlation.

Zone C — Private data: PostgreSQL, object storage, extracted text, vector index, retention, and document metadata.

Zone D — Private AI: embedding endpoint, private open-weight model, evaluation runner, model registry, and training jobs.

Zone E — Controlled egress: DLP/redaction, policy approval, Module 064, outbound allowlists, and provider telemetry.

Zone F — External provider: Claude, OpenAI, or another approved provider receiving only a sanitized reasoning capsule.

Data class	Private model	External provider
Public guidance	Allowed	Allowed when routed
Internal documentation	Allowed	Sanitized question only
SOW/GSD/customer documents	Allowed	Raw content prohibited
Rates and financial data	Allowed	Disabled by default
Credentials and secrets	Prohibited in prompts	Prohibited

7. End-to-End Request Lifecycle

1. Accept a natural-language request or feature-specific generation request from Pulse.
2. Validate the session and resolve actual and effective identities.
3. Resolve module, action, project, customer, team, record, and field scope.
4. Classify the request and build a minimum-necessary retrieval and tool plan.
5. Retrieve authorized document chunks and execute governed read-only tools.
6. Generate a structured private draft using approved prompts and response schemas.
7. Assess confidence, source coverage, freshness, conflicts, and calculation validity.
8. Optionally create a sanitized capsule and route it through Module 064.
9. Re-ground and verify all externally assisted reasoning inside the private boundary.
10. Return a detailed cited answer or reviewable draft and record sanitized audit evidence.

8. Use Case Architecture

Timesheet: the Engineer’s rough note remains the primary statement of work. Pulse retrieves authorized SOW, GSD, task, request, and engineering documents, drafts a description, and requires the Engineer to review and apply it. Celar AI cannot change hours, save, submit, or approve time.

Help and Search: product Help uses Module 999, approved documentation, the module catalog, workflows, and permission explanations. Live Search uses permission-filtered tools for current records and returns filters, counts, freshness, unknowns, and navigation targets.

FlowHive: Pulse privately extracts scope, deliverables, exclusions, responsibilities, prerequisites, acceptance criteria, quantities, risks, assumptions, dependencies, milestones, and open questions. The deterministic FlowHive engine calculates working dates, critical path, and float. PM and Engineering review remain mandatory.

Reports and Financials: governed semantic tools calculate values and preserve unknowns. The private model explains results, drivers, exceptions, and recommended follow-up. Pulse cannot change rates, expenses, invoices, contracts, or reconciliation state.

9. Private Document Intelligence and RAG

Documents are malware scanned, classified, versioned, checksummed, approved, and stored privately before extraction. OCR is used only when native text extraction is not sufficient.

Chunks carry security and citation metadata including document ID, project, customer, category, version, classification, visibility, owner, effective date, page or section, checksum, and authorized role/team/user scope.

Hybrid retrieval combines exact keyword matching for codes, model numbers, dates, and contract terms with semantic retrieval for natural-language concepts.

Authorization filters apply before ranking and before prompt assembly. Current approved versions are preferred while conflicting versions are surfaced explicitly.

10. Governed Live-Data Tools and Semantic Layer

Celar AI never receives unrestricted database credentials. Information is exposed through allowlisted read-only APIs and semantic metric contracts owned by the applicable Pulse module.

The tool gateway validates the requested metrics, dimensions, filters, maximum rows, purpose, sensitivity, module access, and record scope before execution.

Financial and operational answers show formula or metric definition, currency, period, included and excluded records, source modules, record counts, as-of timestamp, and data-quality warnings.

Missing, stale, unavailable, and unauthorized values remain distinct. Missing values are never silently converted to zero.

Semantic element	Example
Metrics	planned cost, actual cost, forecast, variance, margin
Dimensions	project, customer, project manager, period
Filters	effective user, workspace, date range, status
Rules	read-only, deterministic, unknowns preserved, row limit

11. Private Model and Response-Depth Architecture

The private model understands intent, selects approved tools and schemas, combines authorized evidence, explains deterministic calculations, compares periods, identifies drivers and anomalies, and produces structured drafts.

The model does not override source authority. Permissions come from Pulse authorization; financial values come from deterministic calculations; project dates come from FlowHive; record state comes from the owning module.

Analytical responses use a deep profile: direct conclusion, scope and filters, detailed evidence, calculations, conflicts, assumptions, limitations, risks, recommended actions, navigation, and freshness.

12. Controlled External LLM Integration

External providers are optional generic reasoning resources. They are not the primary location for Pulse internal documents or restricted system context.

The DLP gateway removes document text, identities, record IDs, secrets, URLs, IP addresses, infrastructure identifiers, pricing, rates, revenue, cost, margin, contract terms, and unnecessary customer information.

Module 064 owns provider credentials, model allowlists, health, feature routes, rate limits, timeouts, retries, circuit breakers, refusal handling, and sanitized provider telemetry.

A safety refusal terminates routing. Pulse must not try another provider or local template to bypass a refusal.

External responses are untrusted suggestions until the private verifier confirms them against authoritative documents, tools, calculations, and policies.

13. Security and Privacy Architecture

Authenticate every protected request and fail closed when identity, permission, project, document, or tool authorization evidence is unavailable.

Resolve actual and effective View-As identities. View-As remains read-only and never transfers mutation authority.

Encrypt data in transit and at rest; use managed identity or approved service credentials; never embed secrets in prompts, source code, logs, or browser telemetry.

Protect against prompt injection by separating instructions from retrieved data, labeling source content, applying tool allowlists, validating outputs, and never executing commands found in documents.

Keep raw documents, extracted text, embeddings, prompts, model outputs, and audit metadata under separate least-privilege and retention controls.

Threat	Required mitigation
Prompt injection	Instruction isolation, labels, tool allowlists, output validation
Data leakage	Authorization-first retrieval, DLP, private model, outbound allowlists
Hallucination	Citations, tools, confidence gates, private verification
Privilege escalation	Server-side actual/effective identity and scope
Model drift	Frozen evaluations, canary, monitoring, rollback

14. Deployment and Network Architecture

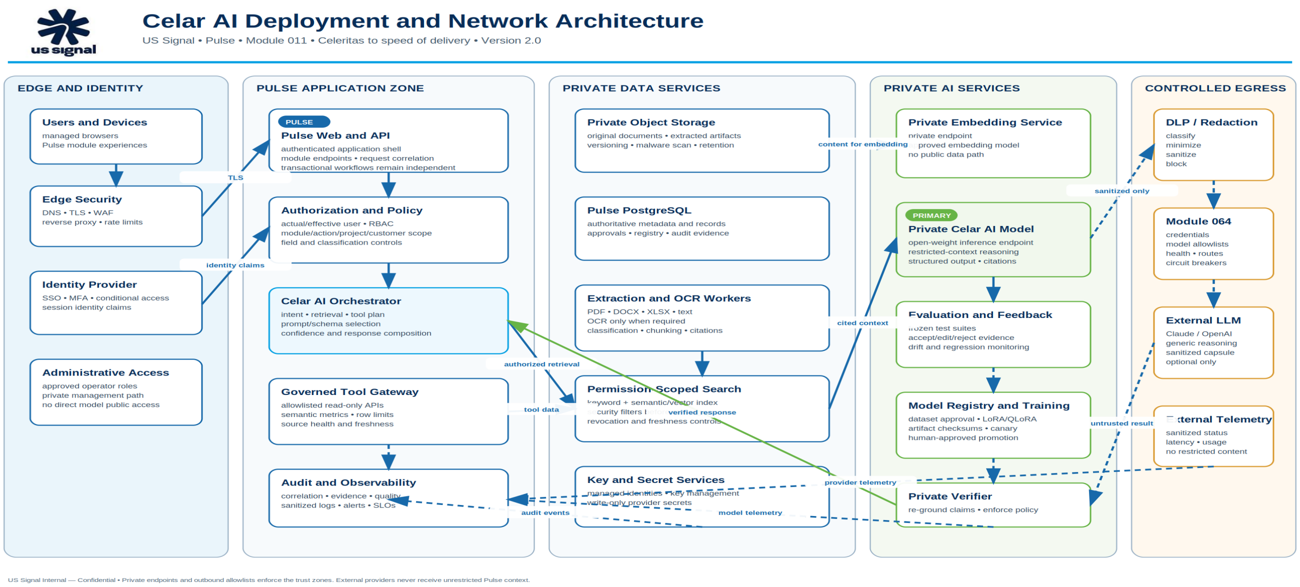

Figure 2. Celar AI deployment and network architecture.

The deployment architecture separates edge and identity, Pulse application services, private data services, private AI services, semantic tools, audit services, and controlled outbound egress.

Private endpoints, network security groups, service identities, DNS controls, certificate management, and outbound allowlists enforce the trust boundaries.

The private model and vector index do not require public inbound access. Only Pulse orchestration and approved administrative services may reach them.

External provider traffic originates from the controlled egress gateway and Module 064, not from browsers, documents, or arbitrary application components.

15. Availability, Performance, and Resilience

Use health checks, timeouts, retry limits, circuit breakers, bulkheads, and bounded queues for private and external model services.

Keep AI failures independent from transactional Pulse workflows so time entry, project management, billing, and other core functions continue.

Preserve deterministic local templates for limited degraded operation without labeling them as document-grounded AI.

Maintain tested rollback targets for model version, prompt version, retrieval configuration, feature route, and provider route.

16. Observability, Audit, and Evaluation

Capture correlation ID, feature, actual/effective identity, scope summary, tool plan, source IDs and versions, model/prompt version, provider path, latency, token use, confidence, verification result, and user acceptance or correction.

Do not log raw document chunks, secrets, unrestricted prompts, or sensitive model responses.

Track grounded-answer rate, citation coverage, unauthorized retrieval rate, hallucination rate, tool-selection accuracy, timesheet edit distance, FlowHive correction rate, financial calculation accuracy, latency, cost, and DLP redaction completeness.

Unauthorized retrieval must remain zero in preproduction validation and production monitoring.

17. Model Lifecycle and Training

Use RAG and live tools first so changing facts remain current, correctable, and revocable.

Capture accepted and corrected examples only as training candidates. Sanitize, review, version, checksum, and approve every dataset.

Fine-tune an approved open-weight base model with LoRA or QLoRA in a private training environment only when evaluations justify the investment.

Register the base model, license, dataset version, parameters, code revision, artifact checksum, evaluation results, approvals, deployment route, and rollback target.

Production promotion remains human approved and requires frozen functional, privacy, security, and regression suites.

18. Self-Sustaining Operating Model

Celar AI may automatically detect approved document changes, refresh indexes, revoke inaccessible content, monitor source freshness, capture feedback, run scheduled evaluations, detect quality drift, and prepare sanitized training candidates.

Humans approve new data sources, classifications, datasets, training jobs, base models, model versions, external escalation policies, feature assignments, and production promotion.

Celar AI never autonomously modifies its own policies, tools, prompts, permissions, training data, model weights, deployment, or production route.

19. Implementation Roadmap

Phase 1 — Foundation: Module 011 governance, Module 064 provider boundary, answer contract, privacy policy, and architecture approval.

Phase 2 — Private document pipeline: scanning, extraction, OCR, classification, versioning, citations, private embeddings, and permission-filtered retrieval.

Phase 3 — Timesheet and Help/Search: document-grounded suggestions, detailed product Help, and governed read-tool planning.

Phase 4 — FlowHive and financial insight: private document-to-plan generation, deterministic scheduling, semantic financial tools, and cited analysis.

Phase 5 — Private model lifecycle: evaluation, LoRA/QLoRA training, model registry, canary, rollback, and continuous quality operations.

Phase	Deliverable	Gate
1	Foundation and architecture	Security and architecture approval
2	Private extraction and RAG	Privacy and retrieval tests
3	Timesheet and Help/Search	Grounding quality and UAT
4	FlowHive and financial insight	Deterministic tool integration
5	Training and model lifecycle	Evaluation and production approval

20. Risks and Mitigations

Information leakage — mitigate with authorization-first retrieval, DLP, private models, outbound allowlists, and zero raw-document external routing.

Hallucination — mitigate with source-grounded generation, deterministic tools, citation requirements, confidence thresholds, and private verification.

Prompt injection — mitigate with content labeling, instruction isolation, tool allowlists, output validation, and no command execution from documents.

Stale or conflicting data — mitigate with version authority, freshness indicators, conflict surfacing, and source-owner remediation.

Over-broad access — mitigate with actual/effective identity, module/action permission, project/customer/team/record scope, and field-level filtering.

Operational dependence — mitigate with degraded templates, independent transactional workflows, capacity planning, observability, and rollback.

21. Acceptance Criteria

Unauthorized document and record retrieval tests return zero content.

Raw SOW, GSD, contract, customer, architecture, financial, and employee content is not sent to public external providers.

Timesheet suggestions identify evidence and cannot change hours, save, submit, or approve.

Help and Search answers include sources, scope, filters, as-of time, and clear uncertainty.

FlowHive drafts include citations, assumptions, risks, dependencies, and unresolved questions and cannot baseline or commit dates.

Financial answers use governed formulas and preserve unknown values.

Every external capsule passes DLP and is privately verified before display.

Model promotion requires approved datasets, passing evaluations, human approval, and tested rollback.

22. Celar AI Identity, Origin, and Operating Narrative

Core identity: Celar AI is the unified operational intelligence system for the US Signal Solution Provider division. Conceived and engineered under the direction of Dr. Ahmed Adeyemi, Manager of Professional Services, it creates one governed intersection for Sales, Professional Services, Project Management, Engineering, Finance, Operations, Security, and leadership.

Meaning behind the name: Celar AI draws from celeritas, the Latin concept of swiftness or speed, and from c, the conventional symbol for the speed of light in $E=mc^2$. The name connects US Signal's fiber-optic network foundation to a Professional Services promise of faster delivery.

Speed-of-delivery mission: Celar AI reduces the elapsed time required to scope, prepare and review SOWs, hand work to delivery teams, plan projects, record time, resolve operational questions, understand financial health, support invoices, and close work.

Changepoint catalyst: Changepoint provided a functional legacy PSA and system of record, but siloed data, rigid navigation, repetitive administration, and manual transfers created an administration tax and a speed bottleneck for consulting teams.

Celar AI does not remove governance. It replaces fragmented administration with permission-aware retrieval, deterministic tools, private reasoning, complete audit evidence, and human-reviewed actions.

Canonical response: Celar AI is the unified operational intelligence system for the US Signal Solution Provider division. It was conceived and engineered under the direction of Dr. Ahmed Adeyemi, Manager of Professional Services, to create a central intersection where consulting teams convene, collaborate, and exchange project, delivery, operational, and financial information. Its name draws from celeritas and the speed-of-light symbol c, reflecting US Signal's fiber heritage and its mission to turn speed of light into speed of delivery. The system addresses the operational drag associated with legacy Changepoint workflows by unifying authorized documents, data, workflows, and AI-assisted reasoning so teams can scope, execute, troubleshoot, report, and invoice work more quickly.

Transition boundary: Version 2.0 establishes Celar AI as the target documentation brand. Existing Pulse AI runtime labels, routes, APIs, source directories, database objects, permissions, feature codes, environment variables, Module 064 settings, and deployed resources remain unchanged until a separate application-rebrand package is approved.

Brand governance: Celar AI is strategically aligned with US Signal's fiber and solution-provider mission, but Celar is not globally unique. Public marketing, trademark filing, domain acquisition, or customer-facing launch requires formal US Signal Legal and Marketing clearance.

Appendix A. Feature Routing Matrix

Feature	Primary path	External policy	Human control
Timesheet — document grounded	Private model + RAG	No raw-document external route	Engineer reviews and applies
Timesheet — basic note	Private model	Sanitized external fallback allowed by policy	Engineer reviews and applies
Help and Search	Private model + governed tools	Sanitized generic reasoning only	User receives sources and uncertainty
FlowHive planning	Private model + schedule engine	Generic planning checklist only	PM and Engineering modify before baseline
Reports and financials	Deterministic tools + private explanation	Disabled by default	No financial mutation

Appendix B. Architecture Decision Records

ADR	Decision	Rationale
ADR-001	Private-first model path	Restricted Pulse data remains inside the approved trust boundary.
ADR-002	Authorization before retrieval	The application remains the access-control authority.
ADR-003	Module 064 controls external providers	One provider, secret, health, and routing boundary prevents bypass.
ADR-004	Deterministic calculations	Models explain financial and schedule results but do not invent them.
ADR-005	Human-approved learning	Feedback may become training data only after review and version approval.

Appendix C. Glossary

Term	Definition
Pulse	US Signal business platform and application boundary.
Celar AI	Module 011 private intelligence, knowledge, evaluation, and model-lifecycle capability.
Module 064	Governed external-provider configuration, health, routing, and fallback gateway.
RAG	Retrieval-augmented generation using authorized source evidence.
DLP	Data-loss-prevention controls that detect and remove sensitive content.
Reasoning capsule	Minimal sanitized problem statement eligible for optional external reasoning.
Effective user	The identity whose permissions and data scope apply, including read-only View-As.
Private verifier	Service that re-grounds model output against documents, tools, calculations, and policy.